

HACKOUT'26

Renewable Energy Intelligence

GRID-AWARE P2P RENEWABLE ENERGY TRADING MARKETPLACE

A network-constrained local energy market for fair pricing, lower losses, and safer distribution-grid operation

Selected Problem Statement

Renewable Energy P2P Trading Marketplace - Households with rooftop solar can sell surplus energy directly to nearby consumers using dynamic pricing based on real-time supply, demand, and grid congestion.

Team: _____

Members: _____

Submission Version: 1.0 | September 2026

1. Executive Summary

The proposed platform is a grid-aware peer-to-peer (P2P) renewable energy marketplace that enables rooftop-solar prosumers to sell verified surplus generation to nearby consumers. Unlike a basic marketplace that matches buyers and sellers only on price, the platform performs market clearing against a digital twin of the local distribution feeder. Every candidate trade is evaluated for economic feasibility, expected losses, feeder congestion, and available network capacity.

The market clears on short time blocks (e.g., 15 minutes) using a network-constrained double-auction mechanism. A dynamic price is then formed from the cleared market price plus scarcity, network-loss, and congestion components. A forecasting layer predicts solar generation and household demand and uses forecast uncertainty to limit how much energy a prosumer can safely commit. Finally, a smart-meter oracle verifies delivered energy, and a smart contract records the trade and automates settlement. This off-chain intelligence + on-chain verification architecture keeps the implementation feasible while making blockchain useful rather than decorative.

Top-3 proposition

We do not simply trade renewable energy peer-to-peer; we trade it only when the grid can physically and economically support the transaction. The result is a market that simultaneously improves prosumer value, consumer price, renewable utilization, and feeder health.

Why this is different

- Network-aware matching: electrical feasibility matters, not just geographic proximity.
- Explainable dynamic pricing: the user can see how market scarcity, losses, and congestion produced the final price.
- Forecast uncertainty: predicted solar surplus is converted into conservative tradable capacity rather than blindly committed.
- Verified delivery: smart-meter readings are authenticated before settlement, connecting physical energy to digital transactions.
- Benchmark-driven demo: performance is compared against grid-only and naive P2P baselines using measurable metrics.

2. Problem Statement & Opportunity

HackOut'26 frames the challenge around rooftop-solar households that may produce surplus electricity during high-generation periods while nearby consumers continue to purchase electricity at higher retail prices. The problem explicitly asks for P2P trading with dynamic pricing tied to supply, demand, and grid congestion, and identifies prosumers, consumers, utilities, and regulators as users/stakeholders.

Stakeholder	Current friction	Platform value
Prosumer	Surplus may be exported at a low feed-in value; generation is intermittent.	Higher-value local sales, forecast-aware commitments, transparent settlement.
Consumer	Retail electricity can be expensive even when local renewable supply exists.	Access to local surplus with a transparent, dynamic price.

Stakeholder	Current friction	Platform value
Utility / DSO	Uncoordinated local trades can increase feeder loading, losses, or operational risk.	Visibility, network constraints, congestion signals, and auditability.
Regulator	Need traceability and confidence in market outcomes.	Verifiable trade records, metering evidence, settlement history, and analytics.

Design objective

Maximize the economic and renewable-energy value of local trades while ensuring that every cleared transaction is compatible with the simulated distribution network. The hackathon prototype should optimize for demonstrable value rather than attempting to reproduce a full utility-scale market.

3. Research & Industry Benchmark

Existing work shows that P2P energy trading is no longer a purely conceptual use case. For example, Powerledger and Tata Power-DDL demonstrated a Delhi pilot in which prosumers and consumers traded solar energy in a dynamic-pricing environment using an auction pool and a blockchain audit trail. Recent academic work goes further by incorporating network constraints, bilateral losses, congestion management, electrical-distance/network-utilization charges, and grid-secure operating envelopes.

Benchmark direction	Observed approach	Implication for our prototype
Industry P2P pilot	Dynamic pricing, auction-pool matching, blockchain audit trail, smart-meter ecosystem.	Blockchain + auction alone is not the differentiator.
Double-auction research	Buyer bids and seller asks are cleared to improve economic efficiency.	Use double auction as the market core, then add network constraints.
Congestion-aware markets	Continuous/periodic trading is constrained by feeder capacity and congestion costs.	Reject or reroute trades that overload the network.
Loss-aware clearing	Bilateral/participant-specific losses can be incorporated into market clearing or settlement.	Price delivery cost rather than just seller ask.
Electrical-distance / network charges	Electrical distance is used to represent network impact more faithfully than simple geographic distance.	Use feeder-path/network sensitivity in matching.
Smart-contract research	Blockchain is used for market operations, auditing, monitoring, and automated settlement.	Keep optimization off-chain; keep commitments, verification and settlement on-chain.

4. Proposed Solution Architecture

Grid-Aware P2P Energy Market - System Architecture

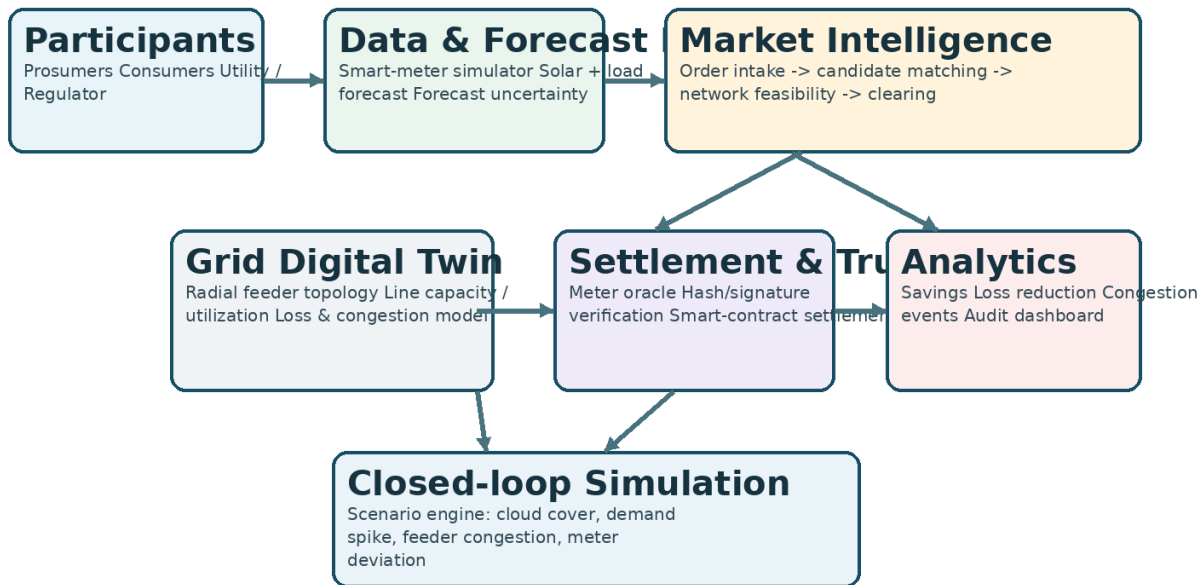


Figure 1. Proposed end-to-end architecture. Optimization and time-series data remain off-chain; trusted commitments and settlement are recorded on-chain.

Core modules

Module	Purpose	Hackathon implementation
Participant & order service	Registers users and receives bids/asks.	REST API + PostgreSQL/SQLite.
Forecast engine	Predicts solar generation and household load.	XGBoost or lightweight time-series model; optional live weather/radiation API.
Market engine	Clears bids/asks for each time block.	Python service using sorting + constrained optimization.
Grid digital twin	Represents feeder nodes, lines, capacities, and flow.	8-15 node radial feeder; linear screening + backward/forward sweep validation.
Dynamic pricing engine	Calculates delivered energy price.	Market price + scarcity + loss + congestion components.
Meter oracle	Authenticates actual delivery.	Signed meter simulator; hash or digest anchored on-chain.
Smart contract	Records commitments and releases settlement.	Solidity on local/private EVM test network.
Web dashboard	Makes the algorithms visible.	React + charting + interactive feeder map.

5. Market-Clearing & Matching Approach

The proposed market operates in short time blocks, such as 15 minutes. Each prosumer submits an offer containing available energy, minimum acceptable price, delivery time block, and network node. Each consumer submits a request containing desired energy, maximum acceptable price, delivery time block, and node.

Network-Aware Market Clearing Loop

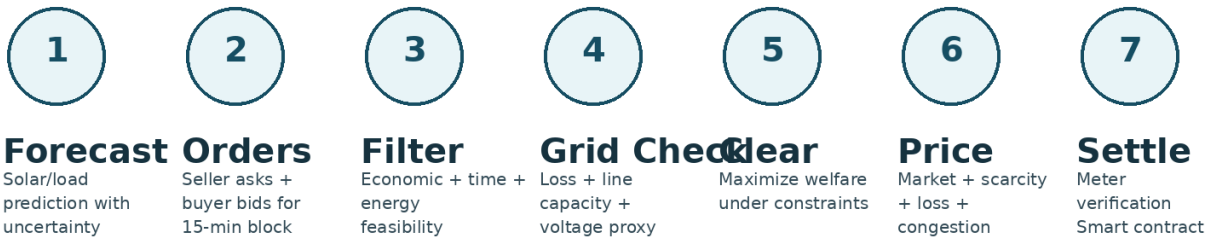


Figure 2. Market-clearing loop used in the prototype.

5.1 Candidate matching

Rather than immediately pairing the highest bid with the lowest ask, the engine constructs feasible seller-buyer pairs for the same time block and then estimates the delivered cost for each pair. A candidate pair is removed if it fails basic economic, temporal, or network checks.

Effective delivered cost

$$C(s,b) = \text{Ask}(s) + \text{LossCost}(s,b) + \text{CongestionCost}(s,b)$$

Buyer surplus

$\text{Surplus}(s,b) = \text{Bid}(b) - C(s,b)$. The clearing engine prefers feasible trades that create positive economic surplus while maximizing total welfare.

5.2 Network-constrained clearing

For the hackathon, the market can be formulated as a small linear optimization problem. Let $q(s,b)$ be energy cleared between seller s and buyer b . The objective is to maximize aggregate economic surplus while enforcing seller capacity, buyer demand, and line-capacity constraints. A compact implementation can use `scipy.optimize.linprog` or `OR-Tools`. The final selected trades are then checked against the feeder simulator.

Simplified optimization objective

Maximize $\sum[(\text{Bid}_b - \text{Ask}_s - \text{NetworkCost}_{s,b}) * q_{s,b}]$

subject to seller supply, buyer demand, $q \geq 0$, and $|\text{Flow}_l| \leq \text{Capacity}_l$ for every feeder line l .

5.3 Electrical distance

Geographic distance alone is not a sufficient proxy for network impact. The prototype therefore uses an electrical-distance score based on the feeder path and configurable line-impact weights. A pair that is geographically close can still be expensive if it traverses a constrained feeder segment; conversely, a slightly more distant customer can be cheaper to serve if the electrical path is lightly loaded.

6. Dynamic Pricing Mechanism

The price engine is designed to be explainable. Every trade receives a market component and explicit network adjustments. This makes price formation understandable to prosumers, consumers, utilities, and judges.

Final price decomposition

$P_{\text{final}} = P_{\text{market}} + P_{\text{scarcity}} + P_{\text{loss}} + P_{\text{congestion}}$

Component	Meaning	Example behaviour
P_{market}	Clearing price produced by the auction.	Higher bids / higher asks move the market-clearing level.
P_{scarcity}	Responds to demand relative to available local supply.	Demand spike -> price increases; surplus solar -> price decreases.
P_{loss}	Reflects expected delivery loss for the seller-buyer path.	Longer / higher-loss paths pay a higher delivery cost.
$P_{\text{congestion}}$	Reflects feeder utilization above a configured threshold.	A heavily loaded line makes trades using it more expensive or infeasible.

A practical congestion term for the prototype is $P_{\text{congestion}} = \lambda_c * \max(0, \text{utilization} - \tau)$, where utilization is $\text{Flow}/\text{Capacity}$ and τ is the congestion threshold. The exact coefficients are configurable and should be calibrated for the demo scenario rather than presented as a regulatory tariff.

7. Forecasting & Uncertainty-Aware Trading

Solar output and household demand are uncertain. A forecast module predicts short-horizon solar generation and consumption from time, weather/radiation, historical values, and site characteristics. A lightweight XGBoost implementation is preferred for hackathon reliability and interpretability; deeper models are unnecessary unless the team already has them.

Conservative commitment

$\text{TradableEnergy} = \max(0, \text{ForecastSurplus} - k * \text{ForecastError})$. This prevents the market from over-committing predicted generation and makes deviation settlement meaningful.

Suggested data strategy

Use a replayable household dataset for repeatable experiments and optionally augment the live demo with a radiation/weather API. Open Power System Data provides household load and solar-generation time series at fine temporal resolution, while Open-Meteo provides satellite/model radiation variables suitable for a live-looking solar forecast pipeline. NREL PVWatts can be used later for PV yield modelling if needed.

8. Blockchain & Smart-Contract Role

Blockchain is deliberately limited to functions where an auditable, tamper-evident record provides value. The optimization engine, forecasts, feeder simulation, and raw time-series measurements remain off-chain. The blockchain stores trade commitments, settlement state, meter-data digests, and relevant audit events.

On-chain	Off-chain
Trade ID and participant references	Raw smart-meter time series
Time block and cleared energy	Forecast models and features
Clearing price / fee summary	Order-book and simulator data
Meter-data hash / verification result	Power-flow calculations
Settlement status and penalties	Dashboard telemetry and charts

8.1 Smart-contract lifecycle

1. Create trade commitment and reserve buyer funds / demo credits.
2. Wait for the delivery period and receive the authenticated meter record.
3. Verify that the meter digest matches the expected record.
4. Calculate delivered-versus-scheduled energy and any deviation adjustment.
5. Release seller settlement, record fees/penalties, and emit an audit event.

Why not put the optimizer on-chain?

Network clearing and time-series computations are iterative and data-heavy. Keeping these off-chain preserves performance and keeps the blockchain focused on verification and settlement - the parts where an immutable shared record is useful.

9. Demonstration Plan

The demo should be an interactive simulation rather than a static CRUD workflow. The central screen is a feeder map plus market metrics, while a scenario controller triggers events that visibly change prices, matches, and network loading.

Demo event	Expected system response	Judge-visible evidence
Sunny / surplus period	Local supply increases; P2P price falls; more trades clear.	Supply-demand chart + seller revenue.

Demo event	Expected system response	Judge-visible evidence
Cloud cover event	Forecast drops; conservative tradable capacity shrinks.	Forecast band + reduced available offers.
EV / demand spike	Local scarcity raises price and changes matches.	Real-time price movement.
Feeder congestion	High-utilization path receives congestion cost or becomes infeasible.	Line turns congested; trade reroutes.
Meter deviation	Actual delivery differs from schedule.	Smart contract settles actual amount and deviation.

Signature moment

Show one buyer who initially selects the cheapest seller. After congestion is introduced, the system rejects that route and rematches the buyer with a slightly more expensive but electrically safer seller. Then explain the new price line-by-line.

9.1 Benchmark comparison

To prove the algorithm matters, the same scenario should be replayed in three modes: (A) grid-only settlement, (B) naive P2P matching by price, and (C) proposed grid-aware P2P. The report and dashboard should show measured differences rather than hand-picked claims.

Metric	Baseline A: Grid-only	Baseline B: Naive P2P	Proposed: Grid-aware P2P
Consumer energy cost	Measure	Measure	Measure
Prosumer revenue	Measure	Measure	Measure
P2P renewable share	0 / baseline	Measure	Measure
Network losses	Measure	Measure	Measure
Peak feeder utilization	Measure	Measure	Measure
Congestion violations	0 by construction	Can occur	Target: 0 after clearing

10. Technology & Implementation Plan

Layer	Recommended technology	Reason
Frontend	React + TypeScript + Map/D3/Recharts	Interactive market and feeder visualization.
Backend API	FastAPI or Node.js/Express	Simple REST interfaces between UI and engines.
Optimization	Python, NumPy, SciPy / OR-Tools	Rapid prototyping of market clearing and constraints.
Forecasting	XGBoost + scikit-learn	Strong tabular baseline with manageable training/inference.

Layer	Recommended technology	Reason
Grid model	Python network simulator; radial feeder equations	Small, controllable digital twin for demo scenarios.
Database	PostgreSQL	Users, orders, trades, telemetry summaries, results.
Blockchain	Solidity on local/private EVM (Hardhat/Ganache-style environment)	Fast testing of contract lifecycle without public-chain cost.
Optional oracle	Backend signer + meter simulator	Bridges simulated physical measurements to settlement.

10.1 Suggested implementation sequence

Phase	Deliverable
Phase 1 - Core market	User model, seller/buyer orders, 15-minute blocks, basic double auction.
Phase 2 - Grid intelligence	Feeder topology, flow/utilization, loss estimate, infeasible-trade rejection.
Phase 3 - Dynamic price	Scarcity + loss + congestion price components and explanation panel.
Phase 4 - Forecast	Solar/load forecast, uncertainty margin, conservative offer capacity.
Phase 5 - Trust	Meter simulator, oracle verification, smart-contract settlement.
Phase 6 - Demo polish	Scenario controls, animation, benchmark comparison, metrics, narrative.

11. Security, Trust & Practical Constraints

The prototype should explicitly acknowledge that physical energy cannot be secured by blockchain alone. Trust enters at the edge - in meter authenticity, participant identity, and the oracle path. The solution therefore uses authenticated meter events, role-based API access, signed data payloads, and immutable trade/settlement records. Raw household consumption data should be minimized on-chain to reduce privacy exposure.

Risk	Prototype control
False meter readings	Signed meter-simulator payloads + hash verification.
Over-commitment from forecast	Uncertainty-adjusted tradable capacity.
Manipulated order data	Authenticated API + server-side validation + audit events.
Grid violation	Network-constrained clearing and post-clear power-flow validation.
Privacy	Store hashes/references on-chain; keep detailed consumption off-chain.

Risk	Prototype control
Blockchain performance	Keep large data and optimization off-chain.

12. Success Metrics

The prototype will be evaluated through measurable scenario outcomes. A strong submission should report at least:

- Average consumer price and percentage savings versus the grid-only baseline.
- Average prosumer revenue per kWh versus the simulated feed-in baseline.
- Local renewable self-consumption / P2P renewable share.
- Total network loss and peak feeder utilization.
- Number of congestion violations before and after network-aware clearing.
- Forecast error and forecast-to-commitment deviation rate.
- Percentage of trades successfully settled against verified meter evidence.

13. Future Scope

A production-grade system could integrate real smart-meter standards, DSO/utility operating envelopes, batteries and EVs as flexible participants, more detailed AC optimal power flow, privacy-preserving market participation, permissioned multi-organization blockchain infrastructure, and regulatory tariff/settlement interfaces. These are intentionally outside the critical hackathon build unless the core platform is already stable.

14. Conclusion

The proposed Renewable Energy P2P Trading Marketplace is designed as a local energy market rather than a simple buy/sell application. Its key contribution is the integration of economic market clearing with the physical constraints of the distribution network. Forecast-aware offers, electrical-distance matching, loss- and congestion-aware pricing, authenticated meter evidence, and automated settlement provide a coherent technical story across the full path from renewable generation to consumer delivery.

Final positioning

Blockchain provides trust and auditability. The real intelligence is the grid-aware market-clearing engine that decides which renewable-energy trades should happen, at what price, and under what network conditions.

15. References & Research Sources

- [1] HackOut'26 - Problem Statements & Guidelines, Renewable Energy Intelligence theme, "Renewable Energy P2P Trading Marketplace." User-provided HackOut'26 problem statement document.
- [2] Powerledger, "Tata Power-DDL, India" - P2P solar trading pilot; dynamic pricing, auction pool, blockchain audit trail, 150 sites / >2 MW PV. <https://powerledger.io/clients/tata-power-ddl-india/>
- [3] Powerledger, "Tata Power-DDL rolls out live peer-to-peer (P2P) solar energy trading" (3 Mar 2021). <https://powerledger.io/media/tata-power-ddl-rolls-out-live-peer-to-peer-p2p-solar-energy-trading-a-first-of-its-kind-pilot-project-in-delhi/>

- [4] ScienceDirect, "Distribution network congestion management considering time sequence of peer-to-peer energy trading" - continuous double auction, congestion cost allocation, 11-bus and IEEE 33-bus validation.
- [5] ScienceDirect, Ullah & Park, "Network-constrained local energy market clearing for peer-to-peer trading with bilateral power losses" (Electric Power Systems Research, 2025) - network-constrained clearing, bilateral losses, electrical-distance/network utilization charge.
- [6] ScienceDirect, "Incentive-compatible double auction for Peer-to-Peer energy trading considering heterogeneous power losses and transaction costs" (Applied Energy, 2025).
- [7] MDPI, "Smart Contract Design in Distributed Energy Systems: A Systematic Review" (Energies, 2023).
- [8] ScienceDirect, "Peer-to-peer energy trading under distribution network constraints with preserving independent nature of agents" (Applied Energy, 2024).
- [9] ScienceDirect, "A framework for prosumer-centric peer-to-peer energy trading using network-secure export-import limits" (Applied Energy, 2024).
- [10] Open Power System Data, Household Data - detailed household load and solar generation time series. https://data.open-power-system-data.org/household_data
- [11] Open-Meteo, Satellite Radiation API - radiation variables and 10-minute satellite data including India coverage. <https://open-meteo.com/en/docs/satellite-radiation-api>
- [12] NREL PVWatts - photovoltaic production modelling resource. <https://pvwatts.nrel.gov/>